AMA3020 - Supplementary

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This document provides some additional or more complete derivations for the theory used in the final report.

Variance of the Monte Carlo estimator and $1/\sqrt{\tau}$ convergence

Recall the estimator

$$\hat{I} = \frac{1}{\tau}\sum_{i=1}^{\tau} h(X_i), \qquad h(x) = \frac{f(x)}{p(x)},$$

where $X_i \sim p(x)$ are i.i.d.

Since expectation is linear,

$$\mathbb{E}[\hat{I}] = I.$$

We now compute the variance. Using independence,

$$\begin{aligned}
\mathrm{Var}(\hat{I}) &= \mathrm{Var}\left(\frac{1}{\tau}\sum_{i=1}^{\tau} h(X_i)\right) && (1)\\
&= \frac{1}{\tau^2}\sum_{i=1}^{\tau} \mathrm{Var}(h(X_i)) && (2)\\
&= \frac{\tau\,\mathrm{Var}(h(X))}{\tau^2} && (3)\\
&= \frac{\mathrm{Var}(h(X))}{\tau}. && (4)
\end{aligned}$$

Therefore the standard deviation (standard error) is

$$\sigma_{\hat{I}} = \sqrt{\mathrm{Var}(\hat{I})} = \frac{\sigma_h}{\sqrt{\tau}}.$$

Hence the Monte Carlo error decreases as

$$\sigma_{\hat{I}} \propto \tau^{-1/2}.$$

Variance in importance sampling

The variance of $\hat{I}$ is

$$\mathrm{Var}(\hat{I}) = \frac{1}{\tau}\mathrm{Var}(h(X)).$$

Now

$$\begin{aligned}
\mathrm{Var}(h(X)) &= \mathbb{E}[h(X)^2] - \mathbb{E}[h(X)]^2 && (5)\\
&= \int_\Omega \left(\frac{f(x)}{p(x)}\right)^2 p(x)\,dx - I^2 && (6)\\
&= \int_\Omega \frac{f(x)^2}{p(x)}\,dx - I^2. && (7)
\end{aligned}$$

Thus

$$\boxed{\mathrm{Var}(\hat{I}) = \frac{1}{\tau}\left(\int_\Omega \frac{f(x)^2}{p(x)}\,dx - I^2\right)}$$

This shows the variance depends strongly on the choice of $p(x)$, and is minimized when $p(x)$ is large where $|f(x)|$ is large, motivating importance sampling.

Boltzmann implies Gaussian

In the canonical ensemble,

$$p(q) \propto e^{-\beta U(q)}, \qquad \beta = \frac{1}{k_B T}.$$

For the harmonic oscillator,

$$U(q) = \sum_{i=1}^{K} \frac{1}{2} k q_i^2.$$

Therefore

$$\begin{aligned} p(q) &\propto \exp\left(-\beta \sum_{i=1}^{K} \frac{1}{2} k q_i^2\right) && (8) \\ &= \prod_{i=1}^{K} \exp\left(-\frac{\beta k}{2} q_i^2\right). && (9) \end{aligned}$$

Compare with a normal density

$$\mathcal{N}(0, \sigma^2): \qquad \phi(q) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{q^2}{2\sigma^2}\right).$$

Matching exponents,

$$\frac{1}{2\sigma^2} = \frac{\beta k}{2} \quad \Rightarrow \quad \sigma^2 = \frac{1}{\beta k} = \frac{k_B T}{k}.$$

Hence

$$q_i \sim \mathcal{N}\left(0, \frac{k_B T}{k}\right).$$

Analytical average energy

Since q_i is Gaussian with variance $\sigma^2 = k_B T / k$,

$$\langle q_i^2 \rangle = \sigma^2 = \frac{k_B T}{k}.$$

The potential energy is

$$U(q) = \sum_{i=1}^{K} \frac{1}{2} k q_i^2.$$

Therefore

$$\begin{aligned} \langle U \rangle &= \sum_{i=1}^{K} \frac{1}{2} k \langle q_i^2 \rangle && (10) \\ &= \sum_{i=1}^{K} \frac{1}{2} k \frac{k_B T}{k} && (11) \\ &= \frac{K}{2} k_B T. && (12) \end{aligned}$$

Metropolis–Hastings acceptance rule

Let $T(x \to x')$ be the transition probability of the Markov chain:

$$T(x \to x') = q(x'|x)A(x \to x'),$$

where q is the proposal distribution and A the acceptance probability.

To sample from $p(x)$ we require detailed balance:

$$p(x)T(x \to x') = p(x')T(x' \to x).$$

Substituting,

$$p(x)q(x'|x)A(x \to x') = p(x')q(x|x')A(x' \to x).$$

Let

$$r = \frac{p(x')q(x|x')}{p(x)q(x'|x)}.$$

Choose

$$A(x \to x') = \min(1, r).$$

Then:

If $r \geq 1$:

$$A(x \to x') = 1, \qquad A(x' \to x) = \frac{1}{r},$$

and detailed balance holds.

If $r < 1$:

$$A(x \to x') = r, \qquad A(x' \to x) = 1,$$

and detailed balance also holds.

Thus the stationary distribution of the Markov chain is $p(x)$.